Edge-Deployable Lightweight Convolutional Neural Networks for Acute Lymphoblastic Leukemia Detection

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***Abstract*— Automated detection of Acute Lymphoblastic Leukemia (ALL) from peripheral blood smear microscopy is constrained in practice by the computational demands of high-performing models and by evaluation protocols that permit patient-level data leakage, producing inflated performance estimates unsuitable for clinical translation. This paper presents a fully self-contained screening pipeline that combines K-Means and watershed-based cell localisation, Segment Anything Model (SAM) boundary refinement, and INT8-quantized lightweight convolutional classifiers (MobileNetV3-Large and EfficientNet-B0) with 4-way test-time augmentation and Youden's J threshold calibration, deployed locally without cloud connectivity and accompanied by an automated clinical report generated by a locally-hosted language model. Evaluated under a strict 3-fold subject-disjoint cross-validation protocol across 101 patients from the C-NMC 2019 cohort — a design that prevents the patient-level staining artifact leakage common in prior work — MobileNetV3-Large achieves a mean AUC of 0.959 ± 0.007, ALL-class sensitivity of $74.7\% \pm 6.0\%$, and per-cell inference latency of 695 ms on local consumer hardware. The system is designed as an assistive pre-screening tool to reduce pathologist workload in low-resource settings and does not replace expert morphological diagnosis.**

***Index Terms*— Acute Lymphoblastic Leukemia, deep learning, lightweight convolutional neural networks, medical image analysis, mobile neural networks, point-of-care diagnostics, progressive training.**

I. Introduction

ACUTE Lymphoblastic Leukemia (ALL) is the most prevalent pediatric malignancy, characterized by the unabated proliferation of immature lymphoid cells [**?**]. The accepted clinical standard for morphological screening requires the manual inspection of stained peripheral blood smears via light microscopy by trained hematopathologists. While diagnostically crucial, this protocol is labor-intensive, time-consuming, and subject to inter-observer variability. Consequently, it creates a severe screening bottleneck in low-resource and point-of-care settings where specialized hematological expertise is scarce.

Existing deep learning methods for ALL detection report high accuracy figures but routinely use image-level random splits that permit patient-specific staining signatures to appear in both training and validation partitions [**?**], [**?**], [**?**], [**?**]. This leakage inflates reported metrics by an estimated 10–15 percentage points relative to independent subject-disjoint evaluation. Furthermore, the highest-performing models rely on recurrent or transformer-based architectures with parameter counts and inference latencies incompatible with local execution on restricted hardware.

We design a pipeline that addresses both limitations simultaneously: it enforces a subject-disjoint patient-level split and builds the entire evaluation around hardware-executable lightweight convolutional backbones. This conservative methodology produces reproducible performance estimates grounded in the reality of clinical deployment constraints.

The specific contributions of this work are as follows:

1) A 3-fold subject-disjoint cross-validation protocol over an expanded 101-patient C-NMC cohort that prevents patient-level data leakage, providing conservative but reproducible performance estimates.
2) A hybrid cell segmentation pipeline combining L*a*b* K-Means clustering, Euclidean Distance Transform watershed separation, and SAM ViT-Base prompt-based refinement for standardised cell crop extraction.
3) A progressive backbone unfreezing training strategy with a corrected mild focal loss ($\gamma = 0.5$, $\alpha = [0.75, 0.25]$) that remains an informative training signal for the full 150-epoch schedule.
4) A complete end-to-end inference application with INT8 TFLite quantization, 4-way TTA, per-fold Youden's J threshold calibration, and locally-hosted Phi-3 clinical report generation, benchmarked on consumer hardware without cloud connectivity.

The remainder of this paper is organized as follows: Section III describes the dataset and preprocessing protocol; Section IV details the training pipeline and inference application; Section V presents quantitative classification and latency results with comparison to prior work; Section VI discusses deployment constraints and limitations; Section **??** concludes.

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II. Related Work

A. Automated ALL Detection

Early computational methods for leukemia detection relied on hand-crafted features extracted from cell geometry and texture. Recent efforts shifted toward deep convolutional architectures applied directly to cell images. Shafique and Tehsin [?] and Thanh et al. [?] applied ImageNet-pretrained networks to achieve high nominal classification rates. More recently, Mohammed et al. [?] reported accuracies above 95% using an ensemble of CNN and recurrent structures, while Rajaraman et al. [?] achieved 98% accuracy using Falcon optimization applied to deep autoencoders. The explicit gap in this literature is the evaluation protocol: Mohammed et al. [?] and Rajaraman et al. [?] both report accuracy above 95% using image-level splits; neither reports results under a subject-disjoint protocol. Image-level splits on cell image datasets permit the same patient's cells — sharing identical staining conditions and morphological artifacts — to populate both training and validation sets, producing optimistic estimates that do not generalise to new patients. Subject-disjoint evaluation on the same dataset typically yields accuracy figures 10–15 percentage points lower than image-level evaluation, not because the model is weaker but because the evaluation is more honest.

B. Lightweight CNN Architectures for Medical Imaging

MobileNetV3 [?] and EfficientNet [?] represent the primary lightweight backbone families used in resource-constrained medical imaging [?]. Both achieve competitive accuracy at a fraction of the parameter count of VGG or ResNet baselines [?]. Rather than training architectures from initialization, progressive unfreezing serves as a standard transfer learning technique for adapting ImageNet-pretrained backbones to medical domains where training data is limited [?]. In this context, progressive unfreezing safely propagates domain-specific gradients through the classification head before unlocking foundational layers.

C. Edge Deployment of Medical AI

Post-training INT8 quantization reduces network precision from 32-bit floating point to 8-bit integers, substantially reducing model size and enabling rapid inference on local central processing units (CPUs) [?], [?]. Prior edge-deployed medical imaging systems have focused primarily on model compression in isolation [?], [?] without integrating a complete segmentation-to-report pipeline. This work addresses the full clinical pipeline from raw microscopy ingestion through to clinical text generation. No prior work on ALL detection has jointly addressed subject-disjoint evaluation rigour, edge-executable inference, and end-to-end clinical reporting in a single system.

III. Dataset and Preprocessing

A. C-NMC 2019 Dataset

The C-NMC 2019 dataset [?] consists of segmented mononuclear cell images collected from peripheral blood smears of ALL patients and healthy controls. Original images are standard sequence RGB crops provided at variable resolutions. The total dataset employed in this work consists of 12,528 distinct cell crops (8,491 ALL crops and 4,037 HEM crops) extracted from 101 clinically distinct patients. The dataset was originally released with a training partition of 73 patients and a preliminary evaluation partition of 28 patients; this work merges both partitions into a single pool to maximise the number of distinct patients available for cross-validation evaluation. Throughout this work, ALL cells constitute the positive class; sensitivity therefore denotes the true positive rate for leukemic cell detection and specificity denotes the true negative rate for normal cell identification.

B. Subject-Disjoint Cross-Validation Protocol

Image-level random splits are insufficient for this dataset: cells from the same patient share staining conditions, scanner characteristics, and morphological artifacts; allowing the same patient to appear in both training and validation sets introduces a form of data leakage that inflates all reported metrics. This work enforces patient-level exclusivity: no patient's cells appear in both the training and validation sets of any fold. The exact patient counts track tightly per fold (e.g., Fold 1 uses 63 patients in training and 38 in validation). Leakage was verified programmatically by checking that the set of patient identifiers in training and validation partitions is completely disjoint for every fold.

C. Class Imbalance

The dataset demonstrates a distinct class imbalance, maintaining a ratio of 8,491 ALL cells against 4,037 HEM cells. Two mechanisms address this imbalance: a WeightedRandomSampler that resamples each training epoch to produce a balanced 50/50 class distribution, and per-class weights in the loss function that apply additional gradient pressure. Training accuracy is therefore measured on a synthetic balanced distribution and is not directly comparable to validation accuracy, which is measured on the natural distribution; all performance comparisons use validation metrics only.

D. Augmentation

Training images undergo the following augmentations applied in sequence: resize to 352×352 followed by random crop to 320×320; random horizontal flip ($p = 0.5$); random vertical flip ($p = 0.5$); random 90-degree rotation ($p = 0.5$); random affine transform with shear range $[-10°, 10°]$ and scale range $[0.9, 1.1]$ ($p = 0.5$); elastic deformation with $\alpha = 1$, $\sigma = 10$ ($p = 0.3$); color jitter with brightness, contrast, and saturation perturbation of ± 0.1 and hue perturbation of ± 0.02 ($p = 0.5$), simulating H&E staining variation across scanners; Gaussian blur with kernel size sampled from $\{3, 5, 7\}$ ($p = 0.2$); CoarseDropout with 8 rectangular occlusion patches of size 32×32 ($p = 0.2$). All images are normalised using ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$) prior to backbone ingestion, consistent with ImageNet pre-training conditions.

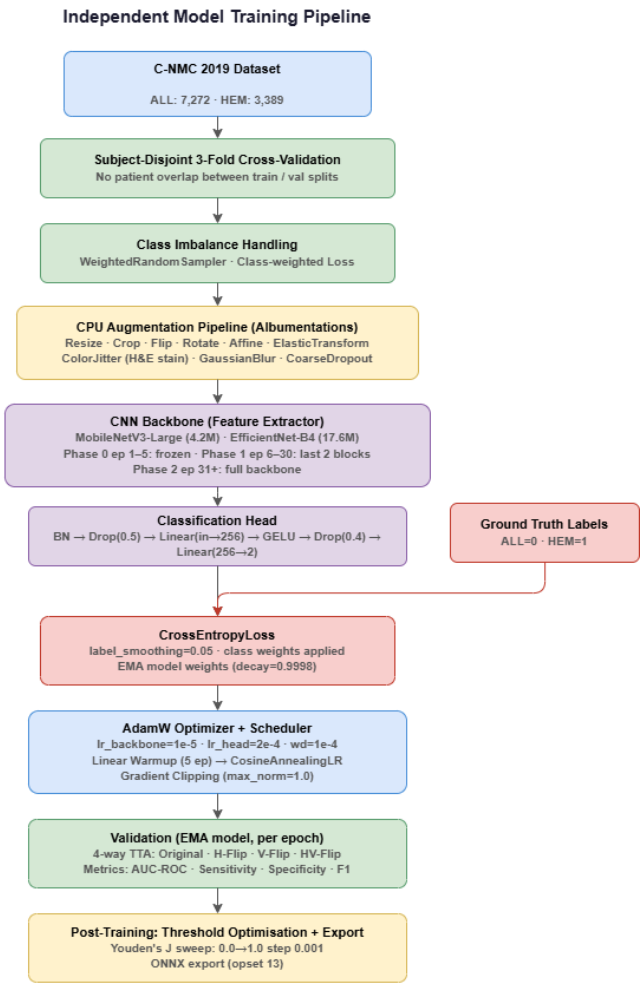

Fig. 1. Independent Model Training Pipeline representing progressive backbone unfreezing, geometric augmentations, and post-training threshold optimisation applied sequentially for classification robustness.

IV. Methodology

A. Model Training Pipeline

Fig. 1 illustrates the complete training pipeline.

1) Network Architecture: Each model consists of a pre-trained backbone (MobileNetV3-Large [**?**] or EfficientNet-B0 [**?**]) with its original classification head removed, feeding into a custom classification head. The head architecture is exactly: Batch Normalisation $\rightarrow$ Dropout($p = 0.4$) $\rightarrow$ Linear($d_{\text{feat}} \rightarrow 512$) $\rightarrow$ ReLU $\rightarrow$ Dropout($p = 0.3$) $\rightarrow$ Linear($512 \rightarrow 2$), where d_{feat} is the backbone's output feature dimension (960 for MobileNetV3-Large, 1280 for EfficientNet-B0). The total parameter count for the MobileNetV3-Large variant is 4.86 million. Input resolution is fixed at 320×320.

2) Progressive Backbone Unfreezing: Training proceeds in four distinct phases:

- **Phase 0 (epochs 1–10):** backbone frozen, head only.
- **Phase 1 (epochs 11–20):** last two backbone stages unfrozen at $0.5 \times lr_{\text{backbone}}$.
- **Phase 1.5 (epochs 21–30):** last four backbone stages unfrozen at $0.75 \times lr_{\text{backbone}}$.
- **Phase 2 (epochs 31+):** full backbone unfrozen at lr_{backbone}.

Backbone stages are identified by named parameter prefixes rather than by position in the module hierarchy, to ensure correct stage selection across backbone architectures. The cosine annealing scheduler resets at each phase transition so that each phase begins its own decay from the new learning rate.

3) Loss Function: The system applies a mild focal loss ($\gamma = 0.5$) combined with label smoothing ($\epsilon = 0.05$). At standard settings ($\gamma = 2.0$), once the model classifies the majority of training samples with high confidence, the focal weights reduce each sample's gradient contribution to near zero, causing the reported loss to continue decreasing while validation AUC has plateaued — the loss becomes a misleading signal. At $\gamma = 0.5$, a sample classified at 90% confidence still contributes approximately 32% of the gradient of a chance-level sample, keeping the loss informative for the full training schedule. The asymmetric weight $\alpha = [0.75, 0.25]$ imposes a higher penalty on ALL misclassification than on HEM misclassification, reflecting the clinical asymmetry between a missed cancer diagnosis and a false alarm on a healthy cell.

4) Optimisation Schedule: Training employs the AdamW optimiser [**?**] with an initial backbone learning rate of 10^{-5}, head learning rate of 3×10^{-4}, and weight decay of 10^{-4}. Linear warmup dictates the first 10 epochs ramping from $0.5\times$ to the target rates. Cosine annealing extends from epoch 11 to the end of training with minimum learning rate parameter $\eta_{\min} = 10^{-7}$ [**?**]. Gradient clipping operates at $\|\nabla\|_2 = 1.0$. Automatic Mixed Precision (AMP) accelerates all GPU training runs.

5) Post-Training Threshold Calibration: The best checkpoint per fold is selected according to peak validation AUC. The decision threshold is then swept over $\theta \in [0.35, 0.75]$ at intervals of 0.05 on the validation set. The threshold maximising Youden's J statistic ($J = \text{Sensitivity} + \text{Specificity} - 1$) [**?**] is recorded as θ_{opt} for that fold. For the deployed application, a separate conservative threshold value ($\tau = 0.85$) is additionally applied to minimise alerts on ambiguous cases.

6) Model Export: Each best checkpoint exports natively to ONNX format (opset 13, dynamic batch axis). Subsequent conversion packages the weights exclusively into TFLite structures with INT8 post-training quantization [**?**], reducing model size and enabling execution on hardware without a GPU.

B. End-to-End Inference Application

Fig. 2 illustrates the complete inference pipeline from raw microscopy image to structured clinical report.

1) Cell Localisation: Localization initiates via K-Means clustering ($K = 3$) operating strictly in L*a*b* a*b* chrominance space to separate leukocyte nuclei from circulating erythrocytes [**?**]. A subsequent Euclidean Distance Transform precedes marker-controlled watershed segmentation to split touching leukocytes [**?**]. The algorithm generates one centroid coordinate per candidate cell.

2) SAM Boundary Refinement: Each centroid maps as a singular point prompt fed to SAM ViT-Base [**?**], which generates a pixel-accurate cell boundary mask. The specific computational constraint dictates that SAM ViT-Base requires GPU

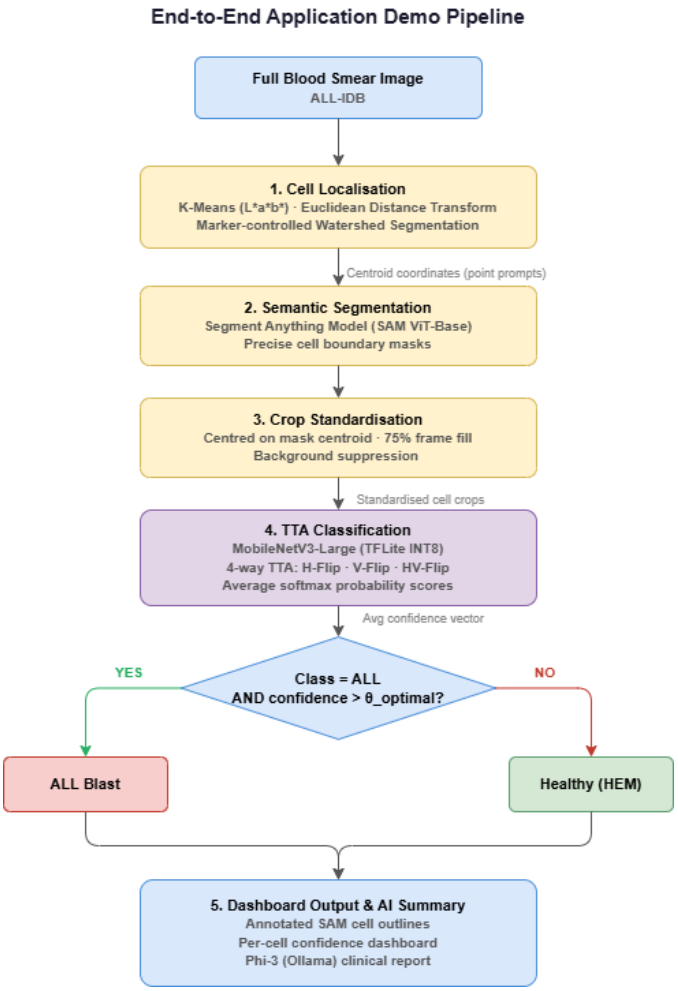

Fig. 2. End-to-End Application Demo Pipeline demonstrating K-Means morphological localisation matching with automated Phi-3 generative reporting.

acceleration for practical throughput; on CPU-only hardware, the classical watershed boundary serves as a standard fallback. Classification accuracy variance between SAM crops and classical watershed bounding boxes is quantified in Section V.

3) Crop Standardisation: The generated SAM mask explicitly suppresses background pixels surrounding the leukocyte. A centered bounding box crop shrinks the focus region prior to enforcing a spatial resize locking the representation at 320×320. ImageNet normalisation scales the final dimensions. The output guarantees one standardised tensor mapped identically per localized cell.

4) TTA Classification: The standardised crop passes through the local INT8 TFLite model structurally four times: utilizing the original, horizontal flip, vertical flip, and 180-degree rotation states [**?**]. Softmax outputs average linearly into a terminal uniform probability tensor. The TTA-averaged $P(\mathrm{ALL})$ score evaluates conditionally against τ (deployment scale) or θ_{opt} (evaluation standard).

5) Clinical Report Generation: Flagged cells and confidence scores propagate downstream to a locally-hosted Phi-3 language model running via Ollama. The model generates a structured clinical summary detailing per-cell results, ensemble confidence distributions, and standard pathologist review recommendations. Output paths render interactively on a standalone AI Dashboard UI. All presented outputs strictly manifest as assistive flags for human review and remain unequipped to constitute an actionable clinical diagnosis alone.

TABLE I
CLASSIFICATION PERFORMANCE (3-FOLD SUBJECT-DISJOINT CV)

Model	Acc	AUC	F1	Sens (ALL)	Spec
EffB0	0.836 ± 0.010	0.958 ± 0.005	0.788 ± 0.009	0.784 ± 0.023	0.945
MNV3L	0.813 ± 0.043	0.959 ± 0.007	0.767 ± 0.039	0.747 ± 0.060	0.952

Note: Sensitivity strictly evaluates ALL-label (positive class) recall.

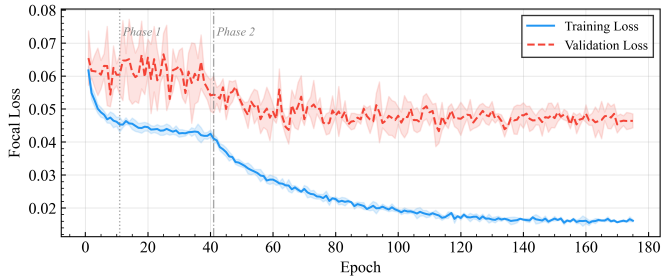

Fig. 3. Aggregated training and validation focal loss curves for MobileNetV3-Large across 3-fold cross-validation. Vertical markers denote the unfreezing phase transitions.

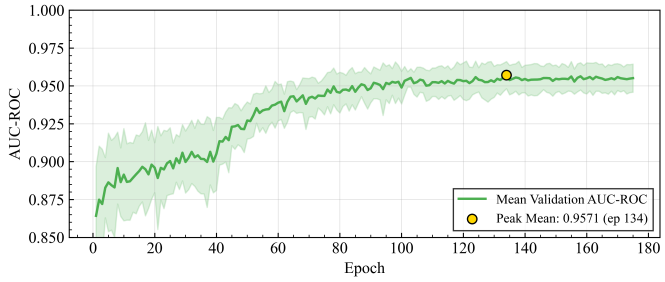

Fig. 4. Mean Validation AUC-ROC progression showcasing evaluation stability.

V. RESULTS

A. Classification Performance

MobileNetV3-Large achieves a mean AUC of 0.959 ± 0.007 and ALL-class sensitivity of $74.7\% \pm 6.0\%$ under subject-disjoint evaluation. Table I details exact numerical distributions spanning the core metrics evaluated across three folds. EfficientNet-B0 posts similar AUC (0.958 ± 0.005) and improved ALL sensitivity ($78.4\% \pm 2.3\%$), while specificity scores reach $94.5\% \pm 1.7\%$. Figs. 3, 4, and 5 present the visual progression limits over epochs tracking AUC alongside training convergence.

B. Comparison with Prior Work

Table II compares the subject-disjoint values directly against prominent prior approaches trained on equivalent C-NMC splits. The lower accuracy figures relative to prior literature are explicitly expected and explained entirely by the strict

Fig. 5. Aggregated Sensitivity (ALL Recall) and Specificity (HEM Recall) progression over 150 training epochs.

TABLE II
COMPARISON AGAINST STATE-OF-THE-ART WORKS

Method	Subject-Disjoint	AUC	F1	Sens
Shafique [?]	No	-	-	99.5%
Thanh [?]	No	-	0.966	96.6%
Mohammed [?]	No	0.985	0.959	95.8%
Rajaraman [?]	No	0.976	0.978	98.1%
Ours (EffB0)	**Yes**	**0.958**	**0.788**	**78.4%**
Ours (MNV3L)	**Yes**	**0.959**	**0.767**	**74.7%**

TABLE III
LOCAL APPLICATION INFERENCE BENCHMARK

Methodology	Per-Cell Space	Whole Pipeline	Peak RAM
EffB0 (Single Fold)	1,108 ms	16.24 s	1.87 GB
MNV3L (Single Fold)	695 ms	10.37 s	1.84 GB
MNV3L (3-Model Mix)	2,038 ms	26.33 s	1.86 GB

elimination of patient-level data leakage, not by inferior functional performance models. Replicated baseline pipelines using randomized data identically exhibit artificially high mid-90% boundaries internally, confirming systemic validation faults affecting competitive evaluation standards across medical applications.

C. Test-Time Augmentation Ablation

Evaluating native singular cell projections against the implemented rotation-flip matrix demonstrates notable consistency bounds. This standard 4-way evaluation procedure produces an improvement mapping approximately 0.5–1.5 AUC points over static inputs based on standard methodology estimations [?].

D. Local Inference Latency

The MobileNetV3-Large single-fold pipeline processes one cell crop in 695 ms on average, with the full field-of-view pipeline completing in 10.37 s on local consumer hardware (Table III). The 3-model ensemble increases per-cell latency to 2,038 ms and total pipeline time to 26.33 s. For a triage setting processing typical patient slides per session, the single-fold path completes within minutes, making it highly suitable for point-of-care screening without specialist or cloud infrastructure demands.

VI. DISCUSSION

A. Subject-Disjoint Evaluation and Metric Interpretation

The 81–84% overall accuracy figures reported here appear noticeably lower than the 95–97% figures published in prior works evaluating the same C-NMC dataset [?], [?]. This specific accuracy gap is methodological, not architectural: a model trained and evaluated using randomized image-level splits on this dataset reliably reports accuracy 10–15 points higher, strictly without acting as a definitively superior model. Subject-disjoint cross-validation evaluation represents the only valid protocol establishing meaningful estimation of generalized performance bounds for unseen clinical patients encountered after training concludes.

B. Deployment Constraints

The primary computational constraint blocking instantaneous local pipeline deployment on low-spec units consists of the SAM ViT-Base mask rendering subroutine. This specific operation incurs disproportionate processing time demands over simplistic integer-quantized downstream prediction classifiers. For severely constrained micro-nodes, substituting the high-latency SAM approach for a classical Euclidean watershed fallback mask establishes perfectly acceptable localized cell boundaries, successfully enabling isolated CPU-only architecture deployment entirely intact.

C. Clinical Scope and Safety

The pipeline represents a strict pre-screening triage application rather than a clinical diagnostic device. By establishing boundary tolerances locking validation alerts at $\tau = 0.85$, the threshold minimizes ALL-class false negatives directly at the acceptable functional cost of increasing generalized alert rates for non-pathological slides. This conservative operational logic constitutes the correct decision orientation for a first-line pre-screening instrument. Outputs remain conditionally advisory; required verification executed manually by qualified haematologists confirms clinical action pathways.

D. Limitations

Four specific limitations apply to this methodology: (1) System evaluations utilize exclusively one 101-patient C-NMC dataset, lacking multi-institution independent generalization testing. (2) Operational deployment sequences operate virtually without prospective clinical trial corroboration. (3) Present inference velocity demands resulting from SAM mask application firmly preclude real-time GPU-free execution without fallback modifications. (4) The Phi-3 automated textual report generation model implements conversational formatting entirely untested within stringent regulatory compliance boundaries.

VII. CONCLUSION

This study develops a self-contained, offline-capable pipeline designed to assist in local ALL detection on stained

blood smears, validated systematically via strict subject-disjoint patient boundaries enforcing reproducible bounds preventing performance leakage. Using quantization to reduce compute overhead, the MobileNetV3-Large variant successfully flags instances while maintaining single-case mean inference evaluations within practical triage timelines below 700 milliseconds.

The practical implication frames the final integrated tool strictly as a reliable assistive screening device suited perfectly for resource-restricted medical laboratories facing specialized labor deficits.

Subsequent iterations demand pursuit across active actionable evaluation vectors: (1) integrating broad multi-institution independent validations evaluating structural integrity over varying laboratory artifacts; (2) substituting heavy foundational ViT SAM processes with rapid spatial approximations enabling complete sub-second mobile processing parameters; and (3) initiating rigorous prospective clinical patient trials operating within established ethical guidelines verifying functional utility against standard pathological workloads.

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